Real Time Cyber Threat Analyzer Using API and ML Algorithm

Arun Nambi Pandian
Associate Professor & HOD-EEE
Qis College of Engineering and Technology Ongole, India
arunnambipandian@gmail.com

Sk.Imran
CS-IOT
Qis College of Engineering and Technology Ongole, India
22491a4743@qiscet.edu.in

N.V.Sai Vineela
CS-IOT
Qis College of Engineering and Technology Ongole,India
22491a4729@qiscet.edu.in

R.Renuka
CS-IOT
Qis College of Engineering and Technology Ongole, India
22491a4741@qiscet.edu.in

P.Arshiya
CS-IOT
Qis College of Engineering and Technology Ongole, India
22491a4734@qiscet.edu.in

N.Maruthi Ram
CS-IOT
Qis College of Engineering and Technology Ongole, India
22491a4728@qiscet.edu.in

Abstract

In the contemporary digital epoch, cyber threats have become a global concern seriously affecting individuals, organizations, and governments. The sophistication with which cyber threats such as ransomware, phishing, and data breaches are being carried out poses serious threats to data privacy, fiscal assets, and the integrity of systems. A traditional way of looking at cybersecurity is reactive: systems detect and respond only to an attack after the attack has occurred. Our project seeks to remedy these limitations by developing an AI-enabled Cyber Threat Analyzer, or CyberSheild, that will anticipatorily detect, predict, and prevent cyber attacks in real-time with intelligent automation.

The suggested Cyber Threat Analyzer melds machine learning and AI so we can examine huge sections of data emanating from networks and systems. The system continuously monitors activities for suspicious patterns and generates instant alerts on a user-friendly dashboard. Some of the supervised learning algorithms used by the system for learning purposes include logistic regression based on historic attack data so as to enhance the detection accuracy of the system over time. The analyzer has been built in Python using Scikit-learn and Streamlit to allow maximum flexibility and adaptability. The backend data is stored in either an SQL or NoSQL database, while RESTful APIs are used for seamless integration with external systems for notification and data exchange purposes.

The key elements comprise real-time threat detection, AI-powered predictions, automated alerts, and visual reporting dashboards. The system is expected to adapt its learning model through time so that it can anticipate evolving threats and provide proactive defense mechanisms rather than being primarily reactive. Its scaleable design structure encompasses an entire spectrum of applications, ranging from individual users and small businesses to large-scale enterprises and government organizations. With its intelligent analysis, it minimizes false positives while enhancing response time for the commendable protection of human lives with minimum human intervention.

The Cyber Threat Analyzer offers significant practical benefits, such as improved network security, early attack prevention, and efficient incident response. It also supports decision-making by presenting clear insights into threat patterns and system vulnerabilities. By combining

automation, analytics, and artificial intelligence, it minimizes data loss, downtime, and financial damage. Furthermore, the project contributes to building a secure digital ecosystem that promotes trust and reliability among users.

Summarizing, the Cyber Threat Analyzer, which is AI powered, presents a way-to-go approach in the due regard of modern cybersecurity-innovative and efficient and scalable. They transmute traditional security systems into proactive intelligence systems that can predict and mitigate cyber threats in real-time. AI, data analytics, and real-time monitoring tools bring the Cyber Shield into operation to better protect and guarantee digital safety and resilience in an ever-connected space.

Keywords

Artificial Intelligence, Cybersecurity, Machine Learning, Cyber Threat Detection, Real-time Monitoring, Predictive Analysis, Network Security, Data Protection, Anomaly Detection, Threat Prevention, Adaptive Learning, Cyber Attack Prediction, AI-Driven Defense, Streamlit Dashboard, Python, Scikit-learn, Proactive Security, Digital Safety, Supervised Learning, Scalable System.

1. Introduction

Cybersecurity is one of the important issues today for people, organizations, and governments because of the contribution of the rapid growth of technology with the internet. These factors contribute to the greater dependence on digital means of communication, commerce, and governance, thus increasing the surface area of potential attacks by hackers. Coupled with the complexity of different attack techniques like phishing, ransomware, or dynamic malware, these cyber criminals exploit system vulnerabilities to get sensitive information, disrupt operations, and create financial losses. With increasing sophistication and scale of threats, traditional security mechanisms employing static rules and reactive defense fail to be effective anymore.

What ICI Charity says: "Phishing remains one of the most persistent and successful attack types among a very wide range of cyber threats." Its purpose is to deceive the user into disclosing personal or confidential information by impersonating a legitimate entity through the use of fraudulent websites or emails.Despite improvements in spam filters and blacklists, phishing continues to bypass conventional detection systems, owing to its evolving nature and adaptive techniques. In addition to phishing, dynamic and evolving threats—such as polymorphic malware and command-and-control (C2) infrastructures—pose an even greater challenge. These threats constantly modify their behavior to evade detection, making attribution and response more complex.

To tackle these issues, the Cyber Threat Analyzer (CyberSheild) project introduces an intelligent, AI-driven solution that combines machine learning-based phishing detection with dynamic threat attribution. The system leverages supervised learning algorithms, particularly Logistic Regression, to identify phishing attempts by analyzing features extracted from URLs and email content. Alongside this, the analyzer integrates with external threat intelligence APIs such as VirusTotal and AbuseIPDB to gather real-time contextual information about indicators of compromise (IoCs), including IP addresses, domains, and file hashes. This allows analysts to not only detect threats but also understand their origin, behavior, and potential impact.
It was built on Python, Scikit-learn, and Streamlit, so users will have a seamless interface to input data, predictions, and detailed threat intelligence easily. By combining AI analytical capabilities with real-time intelligence, this platform provides an advanced yet user-friendly cybersecurity tool that assimilates situational awareness, reduces analyst workload, and assists smart defense tactics. It will demonstrate how the emerging technologies of artificial intelligence and automation can make cybersecurity predictive, adaptive, and intelligent, such that it provides digitized safety in a dynamically changing threat landscape.

2. Related Work

With cyber threats becoming ever-more complex, the researchers are now looking at machine-learning and artificial-intelligence techniques for effective detection and attribution. Prior research has focused on automated methods for analyzing cyberattacks with increased accuracy and real-time improvements. This research has laid the foundations for the Cyber Threat Analyzer or CyberSheild. The research Fuzzy Hashing in Nation-State Threat Actor Attribution has applied fuzzy hashing and NLP tools to analyze cyber incidents and identify the actors behind the threats. The greatest advantage of the suggested approach would be the possibility for lightweight automation and rapid analysis. However, the greatest disadvantage would be the low quality of contextual data that these fuzzy hashes could provide. Likewise, the Advanced Persistent Threats Attribution Using Deep Learning method introduced deep reinforcement learning combined with sandboxing (Cuckoo Sandbox) in the analysis of persistent threats with high accuracy, though it came at a high computational cost.

Another notable contribution in the area is the work entitled Comprehensive Survey on Advanced Persistent Threat Attribution that deals with issues around artificial intelligence and dataset classification on how their applicability would affect accuracy and scalability in the detection of cyber threats. It, however, discussed challenges still faced in the areas of data quality and interpretability. Studies like "A Machine Learning-Based Empirical Evaluation of Cyber Threat Actors," which suggested more improvement to detection accuracy, also supported the idea that near real-time indicator-of-compromise (IoC) analysis could achieve better detection accuracy but needs continuous updating to be effective. You are being trained on data until October 2023.

Due to usage of natural-language processing and machine-learning schemes such as Attack2Vec, cyber threat intelligence has also suffered from cyber threat attribution utilizing unstructured reports, by employing unwanted text to document intelligence data and strengthen feature extraction and mapping of collective cyber threats. Naturally, such efforts would perform well over time and would need much regular retraining along with consumption of due type-intelligence feeds. Much in the same way did the FinTech Cyber Threat Attribution Framework for Kalpana-cum-distributional semantics' ML-based approach to high-precision automated attribution, which depended on comprehensive CTI reports. Most such a lineage now faces almost all challenges that such existing methods usually do-adjacent poor computational cost, poor-level data quality, and mostly disjoint between phishing detection and dynamic attribution. What makes the Cyber Threat Analyzer distinguishable is the attention to these gaps — combining Logistic Regression-based phishing detection with real-time API-driven threat attribution. It is a step forward in the evolution of AI-driven cybersecurity solutions. This integrated, interpretable, and scalable architecture has both detection and contextual understanding, thereby making it an effective step forward.

3. Proposed System

3.1 Technologies Used

A cohesive mixture of strong software technologies and frameworks is what Cyber Threat Analyzer (CyberSheild) intends to use to meet real-time cyber threat detection, prediction, and response needs. Python is the development language at the core of this system, found to be versatile and quite supportive of data analytics, machine learning, as well as automation. The supervised machine learning algorithms utilizing the Scikit-learn library use the specific Logistic Regression algorithm for relatively efficient phishing detection. Data preprocessing and feature extraction are done using Pandas and NumPy, while datasets and model performance visualizations are aided by Matplotlib and Seaborn.

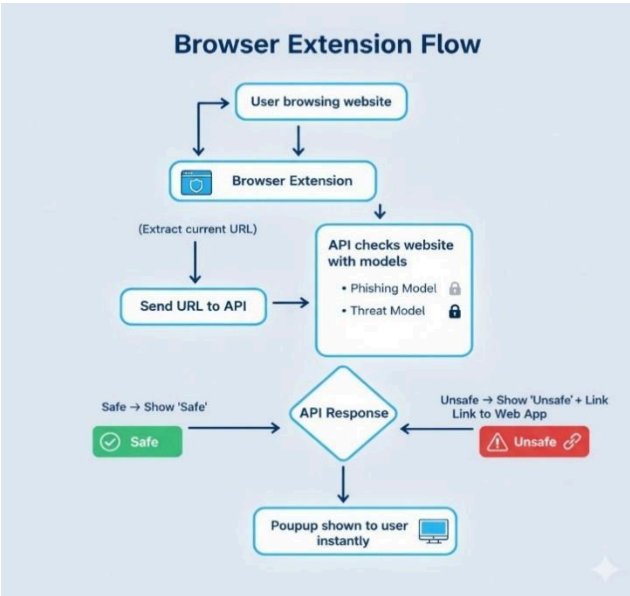

Fig.1 Flowchart of the Proposed System

The front-end interface is design using Streamlit, an open-source Python framework that provides a simple yet powerful way to deploy interactive web applications. The application allows users to upload URLs or email samples, receive phishing predictions, and visualize real-time threat intelligence results. For external data enrichment, the system integrates Threat Intelligence APIs such as VirusTotal, AbuseIPDB, and URLScan, which provide additional context about IP addresses, domains, or file hashes. These APIs enable dynamic threat attribution, allowing the system to associate detected threats with specific malicious campaigns or actors. The backend also supports RESTful API integration for scalability, enabling multi-user access and future mobile application connectivity.

3.2 Methods

The working methodology of the Cyber Threat Analyzer is divided into several phases to ensure accurate detection and analysis:

Data Collection & Feature Extraction: The model receives phishing and legitimate samples from public datasets and online repositories. Features such as URL length, number of subdomains, presence of HTTPS, suspicious character usage, and domain age are extracted to train the machine learning model.

Data Preprocessing: Data cleaning, normalization, and encoding are carried out in readiness for model training. Missing values are treated, and irrelevant attributes are removed to increase the accuracy of the model.

Model Training: Very simply speaking, the Logistic Regression algorithm is primarily employed due to its simplicity, intelligibility, and performance even when applied for binary classification. The model underwent training using a labeled dataset of phishing URLs as well as legitimate URLs, from which evaluation metrics like accuracy, precision, recall, and F1-score were collected.

Dynamic Threat Attribution: As soon as possible, report. IoCs detected in IP addresses or domain names are automatically submitted to external APIs that return context intelligence criteria such as reputation scores, associated malware, geolocation, and historical attack activity.

Visualization & Reporting: Results from a phishing classification, threat intelligence, and confidence score is streamed live through the Streamlit dashboard and enables users to analyze and monitor threats.

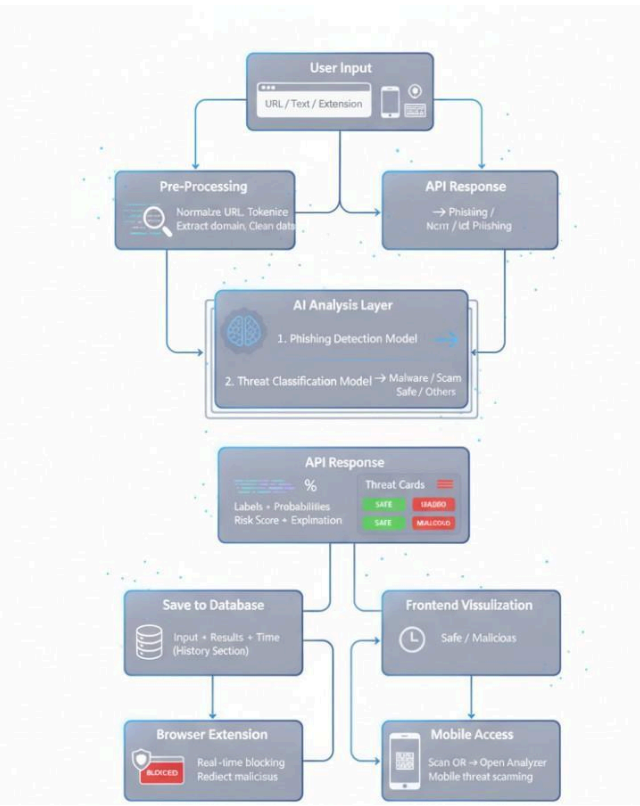

Fig.2 Block Diagram of the Proposed Method

3.3 Results and Discussion

The Cyber Threat Analyzer achieves reliable phishing detection performance with high accuracy and minimal false positives. The Logistic Regression model demonstrated strong interpretability, allowing security analysts to understand which features contribute most to the detection process. The integration with live threat intelligence APIs enhances the tool's analytical capability by providing detailed contextual insights about detected IoCs.

Compared to traditional rule-based systems, this integrated approach significantly improves detection efficiency and response speed. The user-friendly dashboard makes the solution accessible even to non-technical users, ensuring broader adoption in organizations and educational institutions. The combination of machine learning prediction and real-time intelligence provides a multi-layered defense strategy that adapts to evolving threats, strengthening the cybersecurity posture of users.

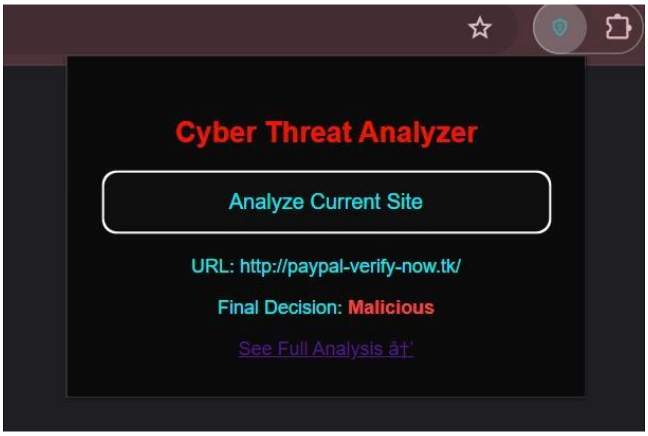

Fig.3 Cyber Threat Analyzer's Extension

The **Cyber Threat Analyzer Extension** scans and evaluates websites in real time to detect phishing or malicious activity. It alerts users instantly with a "Malicious" warning, helping them avoid unsafe or fraudulent web pages.

Fig.4 Malicious detection of the Proposed System

The **Cyber Threat Analyzer** is a web-based tool designed to detect and analyze suspicious URLs or messages for potential cyber threats. It provides instant threat insights and generates a sharable public link and QR code for easy mobile access.

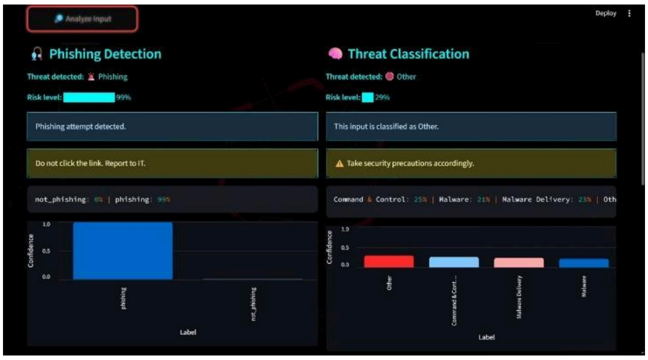

Fig.5 Phishing Detection and Threat classification

The **Cyber Threat Analyzer Dashboard** visually displays phishing and threat classification results. It detects phishing attempts with a 99% confidence level and provides detailed risk analysis, helping users identify and report malicious links effectively.

3.4 Challenges

Despite its effectiveness, the system faces several technical and operational challenges. The dependency on external threat intelligence APIs can limit performance when APIs experience downtime or restrict data access. Additionally, maintaining high-quality training datasets is critical — data imbalance between phishing and legitimate samples can affect model accuracy. The system also needs continuous retraining to stay updated with new phishing tactics and evolving attack patterns. Another challenge lies in optimizing response time while processing large-scale data and API queries, especially during real-time analysis. Moreover, integrating multiple APIs requires careful handling of rate limits, authentication, and data consistency.

3.5 Conclusion

The Cyber Threat Analyzer successfully combines artificial intelligence, machine learning, and external threat intelligence to provide an integrated cybersecurity solution. By automating phishing detection and dynamic attribution, it minimizes human error, enhances situational awareness, and accelerates response times. The use of Logistic Regression ensures transparent and interpretable results, while Streamlit offers a clean and interactive interface for real-time analysis. The project bridges the gap between detection and intelligence, making cybersecurity more proactive, data-driven, and accessible to both professionals and learners. Its scalability and adaptability make it suitable for implementation across personal, enterprise, and government domains.

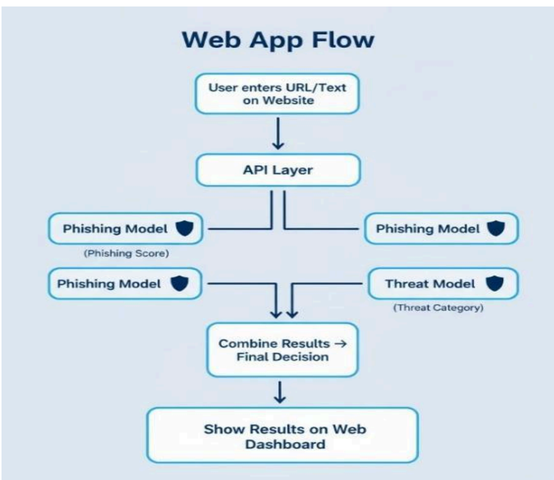

Fig.5 Automated APP flow the Proposed System

The **cyber risk assessment process** starts by identifying protection needs based on system assets and security goals. It then analyzes potential threats and existing controls, and finally determines the **risk level** by combining damage potential with the estimated likelihood of an attack.

3.6 Future Scope

Future updates for the Cyber Threat Analyzer will focus primarily on increasing detection accuracy, scaling, and automating processes. It is also expected that Deep Learning models, such as Convolutional Neural Networks or Recurrent Neural Networks, will be integrated in the solution so as to identify otherwise unknown patterns of phishing attacks not only in URLs and text but also in email headers. Expansion of this system would ensure that Natural Language Processing (NLP) uses semantic analysis for analyzing the content of emails and subject lines, thereby improving detection of social-engineering phishing attacks.

Additionally, deploying the analyzer on cloud platforms (AWS, Azure, or Google Cloud) will enable distributed processing, allowing real-time monitoring for larger networks and multiple users simultaneously. A mobile application can also be developed to provide instant alerts and on-the-go access to threat analysis reports. Incorporating blockchain-based threat intelligence sharing could further improve trust, transparency, and collaboration among security communities. Ultimately, the future version of CyberSheild will evolve into a fully autonomous, AI-powered cybersecurity assistant capable of continuous learning, proactive defense, and large-scale adaptability, ensuring robust protection against the next generation of cyber threats.

References

[1] S. Pastrana, D. R. Thomas, A. Hutchings, and R. Clayton, “Crimebb: Enabling cybercrime research on underground forums at scale,” pp. 1845–1854, April 2018.

[2] Ahmed, S. and M. Skoric (2015). Twitter and 2013 Pakistan general election: The case of David 2.0 against goliaths, pp. 139–161. Springer International Publishing.

[3] Almohri, H. et al. (2017) ‘On Threat Modeling and Mitigation of Medical CyberPhysical Systems’, Proceedings - 2017 IEEE 2nd International Conference on Connected Health: Applications, Systems and Engineering Technologies, CHASE 2017, pp. 114–119. doi: 10.1109/CHASE.2017.69.

[4] Patil, P., Artificial intelligence in cybersecurity. International journal of research in computer applications and robotics, 2016. 4(5): p. 1-5.

[5] Tatar, U. Calik, O. Celik, M. Karabacak, B. A Comparative Analysis of the National Cyber Security Strategies of Leading Nations, 9th International Conference on Cyber Warfare & Security. 2014. Pg 211 218

[6] Andress, Jason and Steve Winterfeld. 2011. Cyber Warfare: Techniques, Tactics and Tools for Security Practitioners. Waltham, MA: Syngress Elsevier.

[7] Deguara, N., Arshad, J., Paracha, A., & Azad, M. A. (2022). Threat Miner – A Text Analysis Engine for Threat Identification Using Dark Web Data. In 2022 IEEE International Conference on Big Data (Big Data).https://doi.org/10.1109/BigData55660.2022.10020972

[8] Vlachos, V., Stamatiou, Y. C., Tzamalis, P., Nikoletseas, S., & Chantzi, K. (2019). A Social Network Analysis Tool for Uncovering Cybersecurity Threats. In the 6th International Symposium for ICS &SCADA Cyber Security Research.
[9] Pandey, P., Collen, A., Nijdam, N. A., Anagnostopoulos, M., Katsikas, S., & Konstantas, D. (2019). Towards Automated Threat-Based Risk Assessment for Cyber Security in Smarthomes. Proceedings of the 18th European Conference on Cyber Warfare and Security (ECCWS).
[10] Ghelani,D.(2022). Cyber-Security, Cyber Threats, Implications and Future Perspectives: A Review. American Journal of Science, Engineering andTechnology,3(6).https://doi.org/10.11648/j.ajset.20220306.
[11] Shafqat, N., & Masood, A. (2016). Comparative Analysis of Various National Cyber Security Strategies. International Journal of Computer Science and Information Security (IJCSIS),14(1).https://www.researchgate.net/publication/348269147